Executive Technology Guide

Inside the HurModel Revolution

How Sapiens Technology® is rewriting the rules of Artificial Intelligence

Classification: General Overview for Business and Innovation (General Public)

Summary

This document presents, in a simple and direct way, the innovative *HurModel*, the definitive architecture created by Sapiens Technology® that solves the major bottlenecks of cost and slowness in current Artificial Intelligence. Imagine being able to teach an AI without it having to make millions of errors to learn. By combining technologies that capture the "big picture" instantaneously, our system eliminates the old need for trial-and-error learning. The result? A proven leap where the AI learns **15 times faster** from the start, and can update itself with new data at a speed **71 times greater**. Next, we explain with simple analogies how we achieve this undeniable superiority.

1 The Magic of HurNet: Understanding the Whole at Once

The great secret of the HurModel lies in replacing the "rigid" way traditional AIs read information (processing one data point at a time) with a system that sees how things connect immediately.

1.1 The Symphony Orchestra Analogy (Augmented Transformation)

Think of a traditional AI as a lay listener trying to understand a song by listening to one instrument at a time: first they focus only on the violin, then only on the piano, to finally try to guess the melody. Our **HurNet** operator acts like an *Experienced Maestro*. When looking at the score, they don't just see isolated notes; they instantly understand the **harmony** among all instruments interacting together at once.

Instead of complex equations, we can summarize our central logic (what we call Deterministic Injection) as follows:

$$\text{Instantaneous Understanding} = \text{Individual Notes} + (\text{Global Harmony among all Elements}) \tag{1}$$

By injecting this "vision of the whole", the AI skips processing steps that previously took days. It already understands the "music" right in the first measure.

Practical Guarantee 1.1 (Direct Learning). In practice, this ensures that our model understands highly complex and nuanced scenarios in a single "leap" of reasoning. It is like reading the perfect summary of a book instead of reading pages and pages of a dictionary, drastically reducing the size and complexity required for the AI to be intelligent.

2 The Direct Route: Eliminating Trial and Error

The current AI industry uses a method called "Gradient Descent" to train models. Imagine a blindfolded person at the top of a mountain, trying to descend by taking small steps and feeling the ground to sense the slope. If they make a mistake, they go back and try another path. This is slow, wastes a lot of energy and money.

The HurModel abandons this blind walk.

2.1 The Mathematical "GPS" (Strict Solution)

Instead of walking blindfolded, we use a mathematical shortcut (which engineers call Pseudo-Inverse). It is as if we handed a **drone with GPS** to the AI. The model simply looks at the entire map, sees where the base of the mountain is, and jumps straight there.

$$\text{Perfect Path} = \text{Calculation of Direct Route (No Random Walking)} \tag{2}$$

This means that the model arrives at the right answer much faster, stably, and without the risk of "getting lost" on the way (which in engineering we call immunity to explosions in the calculation space).

3 Initialization Dynamics: Starting Ahead

When an athlete is going to run the 100-meter dash, they need a long warm-up period. Traditional AIs also suffer from this: the beginning of training is chaotic and slow until the machine "warms up" and begins to understand what it is doing.

3.1 Smart Seeding

We solve this with *Zero-Gradient* Initialization. Imagine if our athlete could start the race already at the 50-meter mark, fully warmed up and at maximum speed.

$$\text{Starting Point} = \text{Prior Analysis} \rightarrow \text{Direct Placement on the Final Stretch} \tag{3}$$

Before the official training even begins, our mechanics quickly study the data and deposit the AI's brain exactly at the ideal point. This doubles efficiency right in the first minutes of use.

4 Unmatched Speed and Energy Savings

The true impact of the HurModel architecture shines when we look at the time and money saved, breaking the limits of what was thought possible in technology.

4.1 Instant Updates (The New *Fine-Tuning*)

Fine-tuning is the process of taking an AI that is already intelligent and teaching it something very specific (like your company's rules).

In old models, teaching a new rule is like asking someone to read an entire encyclopedia again just to memorize a new page. With the HurModel, we eradicate this need. Updating becomes like changing just one "index card in a file", without touching the rest.

$$\text{Update Time (HurModel)} = \text{Focus Only on New Data (Stunning 71x faster)} \tag{4}$$

Operational Supremacy 4.1 (Green and Accessible AI). Since we don't need to "rewrite the encyclopedia" and we reduce mathematical operations to a tiny fraction of the normal amount, the HurModel requires much less electricity and servers. **Sapiens Technology®** proves here that it is possible to run super-powerful AIs on simpler computers (restricted hardware), setting a new ecological and economic standard for the global market.